

ESSENTIALS OF COMPUTING SCIENCES

Project Administration

4th Edition

ESSENTIALS OF COMPUTING SCIENCES

PROJECT ADMINISTRATION
4th Edition

Editors

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Zolidah Kasiran



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Preface

In the Universiti Teknologi MARA Malaysia, the Bachelor Degree programmes of the Computing Sciences discipline typically end with a final year project. This final year project is the capstone of each Bachelor Degree programme as it builds and tests the knowledge and skills acquired throughout the scholastic sessions. In addition, this project contributes as an essential part of the training towards becoming a professional. This book presents the essentials of the Computing Sciences project administration. Its aim is to meet the administrative formalities and procedural needs of students who are embarking on the formulation stage of their final year project, through to project completion. Several categories of readers may benefit from the content of this book as it provides important information pertaining to the Computing Sciences department requirements. The categories include the student who undertakes the project, the supervisor who supervises the project, the examiner of the project, the lecturer who conducts the project courses, and the coordinator who is responsible for managing the project exhibition for the Computing Sciences discipline.

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List of Abbreviations

FYP	Final Year Project
CSP600	Course code for Project Formulation
CSP650	Course code for Project
MAF	Mutual Acceptance Form
FAQ	Frequently Asked Question
ATP	<i>Amanah Tugas Pensyarah</i>
OBE	Outcome Based Education
RES	Result Examination System
SIMS	Student Information Management System

List of Forms and Evaluation Rubrics

F1	Mutual Acceptance Form
F2	Project Motivation Form
F3	Literature Review Evaluation Form
F4	Methodology Evaluation Form
F5	Proposal/Project In-Progress Form
F6(a)	Project Formulation Report Submission Form
F6(b)	Project Report Submission Form
F7	Project Formulation Presentation Form
F8	Project Formulation Evaluation Form
F9	Project Progress Project Presentation Form
F10	Final Project Presentation Form
F11	Final Report Evaluation Form
F12	Project Confirmation of Correction Form
F13	Business Model Evaluation Form
F14	Special Evaluation Qualification Form
F15	Special Evaluation Presentation Form
F16	Special Evaluation Report Form

Chapter 1

Aim

To provide an overview of the Bachelor Degree of Computing Sciences final year project.

1.1 Introduction

A final year project is different from a traditional course in terms of its size, its goals, and its form of supervision, communication and examination. It represents a substantially larger workload than that expected of a single course. While a single course traditionally focuses on the student acquiring knowledge in a specific subject area through lectures and laboratory work, the final year project focuses on deepening the student's understanding of a subject area. Moreover, it provides the student with the training to conduct the project independently using a sound and structured approach.

For a successful completion of a Bachelor Degree in the Computing Sciences discipline, the student must complete the required courses, prepare a project proposal for examination and conduct research work that culminates in an examination of a final year project report with an oral presentation of the results. It is to this capstone of a Bachelor Degree programme that the final year project cultivates the development of a student towards becoming an independent and critical-thinking individual.

1.2 Objective

The book focuses on the final year project within the Computing Sciences discipline. It provides informative materials to both the lecturer and the student in planning, organizing and managing the project towards its successful completion. The contents of the book enable the lecturer and the student to comprehend the administrative process and to apply the report guidelines involved in undertaking a project of this nature.

1.3 Structure

The book is structured into two main sections: (i) the formulation and submission of the project proposal, and (ii) the execution of the proposed project towards completion. For each section, the administrative process and workflow are provided for both the lecturer and the student. The project examination criteria can be discerned from the forms and rubrics that are provided in the appendices as a guide for the lecturer and the student. In addition, the forms provided in the appendices are detachable for use.

Chapter 2

Aim

To provide the administrative process and workflow on the formalities and procedural needs of the Bachelor Degree of Computing Sciences final year project (FYP).

2.1 Project Formulation and Project Administration Process

There are two main sections in this chapter: Project Formulation and Project. The Project Formulation section pertains to a course, of the same name, undertaken by the students in the fifth semester of their undergraduate programme. The Project section relates to a course, of the same name, undertaken by the students in their sixth semester of their bachelor programme. The roles and responsibilities of the lecturer and student are explained herewith.

2.2 Process for the Lecturer

Table 2.1 shows the responsibilities and actions undertaken by the lecturer during the Project Formulation and Project course.

Table 2.1 Administration Process for the Lecturer During the Project Formulation and Project Course

Project Formulation	
Responsibility	Action
Project Formulation Lecturer	1. Explain the following core course structure to the students: • Roles of student, supervisor / Co-Supervisor, examiner, Project Formulation lecturer • Purpose of forms • Evaluation rubrics • Division of marks/assessment <i>(Refer to appendix N for Selection of Supervisors)</i> 2. Acquire the student list according to the program. 3. Provide instructions on the preparation of the project proposal. 4. Provide a list of previous project titles for student reference. 5. Provide a list of the lecturers' areas of specialization. 6. Inform the students to seek potential supervisors. 7. Instruct the students to complete and submit the following forms: • Mutual Acceptance Form (F1) • Project Motivation Evaluation Form (F2)
Supervisor/Co-Supervisor	8. Discuss and confirm the project title with the potential student.

Table 2.1 (Continued)

Responsibility	Action
Supervisor/Co-Supervisor	9. Guide the student in preparing the project proposal. 10. Instruct the student to do the amendments (if any). 11. Check and sign the Proposal/Project In-Progress Form (F5).
Project Formulation Lecturer	12. Instruct the students to complete and submit the following forms: • Literature Review Evaluation Form (F3) • Methodology Evaluation Form (F4) 13. Conduct a session on Research Ethic 14. Evaluate Literature Review Evaluation Form (F3) and Methodology Evaluation Form (F4). 15. Monitor the student's project progress throughout the semester with the Proposal/Project In-Progress Form (F5). 16. Create a student group account with an anti-plagiarism software. 17. Instruct the students to screen the project formulation report with the anti-plagiarism software. 18. Instruct the students to submit the Project Formulation Report and the Project Formulation Report Submission Form (F6(a)).
Project Formulation Lecturer	19. Assign examiner for each student accordingly 20. Plan Project Formulation Presentation session 21. Distribute the proposal reports with the Project Formulation Report Evaluation Form (F8) to the supervisors and examiners.
Project Formulation Lecturer, Supervisor & Examiner	22. Evaluate the proposal through presentation using the Project Formulation Presentation Form (F7)
Supervisor & Examiner	23. Evaluate the report using Project Formulation Report Evaluation Form (F8)
Lecturer	24. Instruct the students to do the amendments (if any). 25. Instruct students to submit REC Forms as appropriate with the project proposal.
Supervisor & Examiner	26. Submit the Project Formulation Presentation Form (F7) and the Project Formulation Report Evaluation Form (F8) to the Project Formulation lecturer.
Project Formulation Lecturer	27. Instruct the students to submit the amended proposal report. 28. Record and upload the Project Formulation course marks into the Result Examination System (RES). 29. Compile and submit the items listed below to the Project lecturer for the following semester: • The student project list • The amended proposal reports

Table 2.1 (continued)

Project	
Responsibility	Action
Project Lecturer	30. Explain the following core course structure to the students: • Roles of student, supervisor, Co-Supervisor, examiner, Project Lecturer • Forms used • Evaluation rubrics • Division of marks/assessment <i>(Refer to appendix N for Roles of Supervisors)</i> 31. Acquire the student list according to the program. 32. Provide instructions on the preparation of the FYP report according to the report guidelines.
Supervisor/Co-Supervisor	33. Provide continued supervision on the student project. 34. Guide the student in preparing the FYP report.
Project Lecturer	35. Monitor the student's project progress throughout the semester using the Proposal/Project In-Progress Form (F5). 36. Instruct student to develop Lean Model Canvas (LMC) <i>(Refer to Appendix O)</i> 37. Create student group account for anti-plagiarism detection software/system. 38. Evaluate the submitted LMC using the LMC Evaluation Form (F13). 39. Evaluate the project progress through presentation using the Project Progress Presentation Form (F9). 40. Instruct the student to screen the FYP report with the anti-plagiarism software/system.
Project Lecturer	41. Update the list of supervisors and examiners. 42. Instruct the students to submit the FYP report with the Report Submission Form (F6(b)). 43. Distribute the FYP reports, with the Final Project Presentation Form F(10) and the Project Report Evaluation Form (F11) to the supervisors and examiners.
Resource Person / Lecturer in Charge	44. Plan and organize the FYP exhibition.

Table 2.1 (Continued)

Responsibility	Action
Resource Person / Lecturer in Charge	45. Brief the students on the FYP exhibition.
Supervisor & Examiner	46. Evaluate the FYP presentation, report and poster using the Final Project Presentation Form (F10) and the Project Report Evaluation Form (F11) . 47. Submit the forms to the Project lecturer. 48. Instruct the students to do the report amendments (if any) according to the report guidelines. 49. Check the amended report and sign the Confirmation of Correction Form (F12).
Supervisor	50. Sign the Supervisor Approval page in the student's final report.
Project Lecturer	51. Collect the following items from each student: • FYP report (including abstract and appendices) (in .pdf and .doc) • Presentation slides • All files related to the project development • Poster • Raw data (If Relevant) • System with test data (If Relevant) • Instructions on System Setup • .apk file / .exe file (If Relevant) 52. Record and upload the Project course marks into the Result Examination System (RES).
Lecturer	53. Submit a set of the FYP reports to the faculty's Academic Affairs Office for archive as resource reference.

2.1.1 Process Workflow for the Lecturer

Table 2.2 shows the responsibilities, workflow, and work processes undertaken by the lecturer during the Project Formulation and Project course.

Table 2.2 Administration Workflow for the Lecturer During the Project Formulation and Project Course

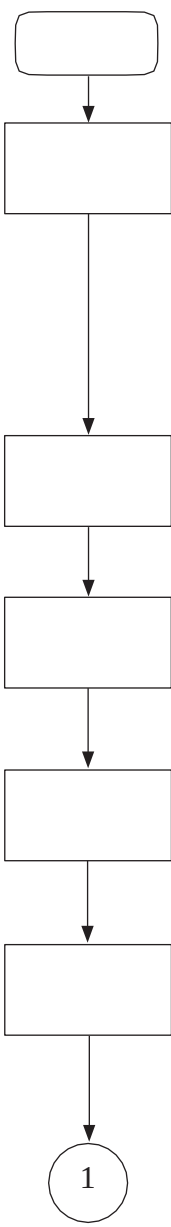
Project Formulation			
Responsibility	Workflow	Work Process	Source/Record
Project Formulation Lecturer		Start	
		Explain the following core course structure to the students:	Project Formulation course – Outcome Based Education (OBE) documents
		• Roles of student, supervisor, examiner, Project Formulation lecturer • Purpose of forms • Evaluation rubrics • Division of marks/assessment	
		Acquire the student list according to the program.	Student Information Management System (SIMS)
		Provide instructions on the preparation of the project proposal.	
		Provide a list of previous project titles for student reference.	FYP coordinator, previous Project Formulation lecturer's teaching portfolio
		Provide a list of the lecturers' areas of specialization.	Faculty website

Table 2.2 (Continued)

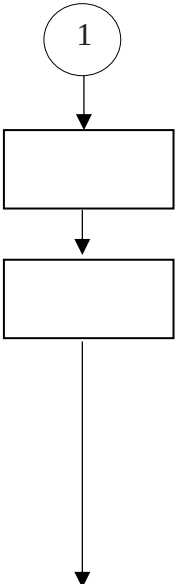
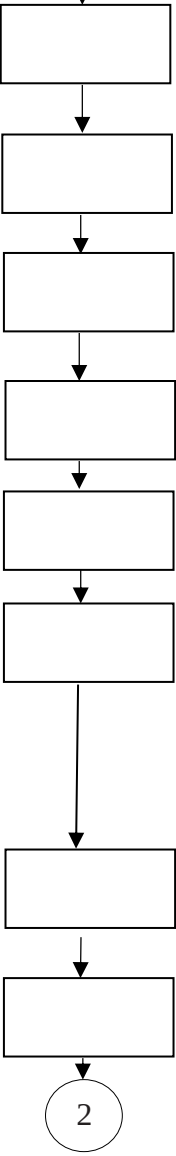
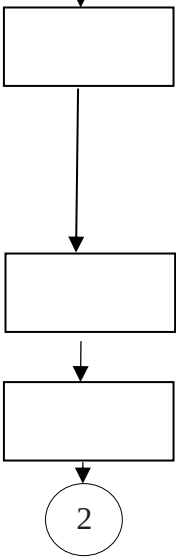
Responsibility	Workflow	Work Process	Source/Record
Project Formulation Lecturer		Inform the students to seek potential supervisors. Instruct the students to complete and submit the F1 and F2 form.	Mutual Acceptance Form (F1) - 3 copies: Project Formulation lecturer, supervisor and student Project Motivation Evaluation Form (F2)
Supervisor		Discuss and confirm the project title with the potential student. Confirm acceptance of the supervision and sign the F1 form. Guide the student in preparing the project proposal. Instruct the student to do the amendments (if any). Check and sign the F5 form.	Mutual Acceptance Form (F1) Proposal/Project In-Progress Form (F5)
Project Formulation Lecturer		Instruct the students to complete and submit the following forms: - Literature Review Evaluation Form (F3) - Methodology Evaluation Form (F4) Conduct a session on Research Monitor the student's project progress throughout the semester with the F5 form.	Literature Review Evaluation Form (F3) and Methodology Evaluation Form (F4) Proposal/Project In-Progress Form (F5)

Table 2.2 (Continued)

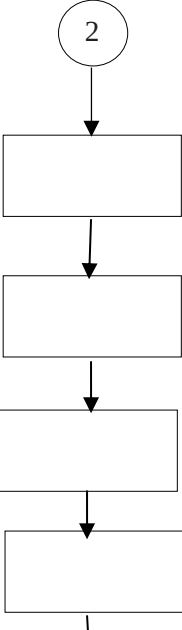
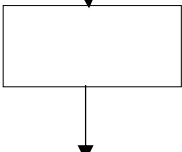
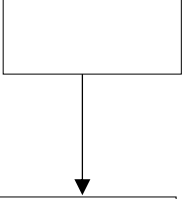
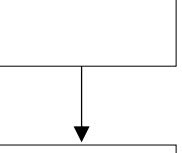
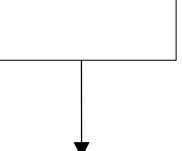
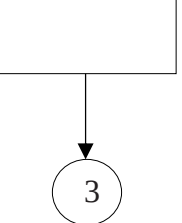
Responsibility	Workflow	Work Process	Source/Record
Project Formulation Lecturer		Instruct the students to screen the proposal report with the anti-plagiarism software. Instruct the students to submit copies of the proposal report and the F6(a) form. Assign examiner for each student accordingly Plan Project Formulation Presentation Session	Anti-plagiarism software Proposal report Project Formulation Submission Form (F6(a))
Project Formulation Lecturer		Distribute the proposal reports with the F8 form to the supervisors and examiners.	Project Formulation Report Evaluation Form (F8)
Project Formulation Lecturer, Supervisor & Examiner		Evaluate the proposal through presentation using the F7 form.	Project Formulation Presentation Form (F7)
Supervisor & Examiner		Evaluate the report using Project Formulation Report Evaluation Form (F8)	Project Formulation Report Evaluation Form (F8)
Project Formulation Lecturer		Instruct the students to do the amendments (if any).	
Supervisor & Examiner		Submit the F7 and F8 forms to the Project Formulation lecturer.	Project Formulation Presentation Form (F7) Project Formulation Report Evaluation Form (F8)

Table 2.2 (Continued)

Responsibility	Workflow	Work Process	Source/Record
Project Formulation Lecturer	<pre> graph TD 3((3)) --> B1[] B1 --> B2[] B2 --> B3[] B3 --> 4((4)) </pre>	Instruct the students to submit the amended proposal report. Record and upload the Project Formulation course marks into the Result Examination System (RES). Compile and submit the items listed below to the Project lecturer for the following semester: • The student project list • The amended proposal reports	Result Examination System (RES) Teaching portfolio

Table 2.2 (Continued)

Project			
Responsibility	Workflow	Work Process	Source/Record
Project Lecturer	<pre> graph TD 4((4)) --> 4.1[] 4.1 --> 4.2[] 4.2 --> 4.3[] </pre>	Explain the following core course structure to the students: • Roles of student, supervisor, examiner, Project lecturer • Forms to be used • Evaluation rubrics • Division of marks/assessment (Refer to appendix N for Roles of Supervisors)	Project course – Outcome Based Education (OBE) documents
		Acquire the student list according to the program.	SIMS
		Provide instructions on the preparation of the FYP report according to the report guidelines.	Appendices in Guidebook
Supervisor	<pre> graph TD 4.4[] --> 4.5[] </pre>	Provide continued supervision on the student project.	
		Guide the student in preparing the FYP report.	FYP Report Checklist (Appendix M)
Project Lecturer	<pre> graph TD 4.6[] --> 4.7[] 4.7 --> 4.8[] 4.8 --> 5((5)) </pre>	Monitor the student's project progress throughout the semester using the F5 form.	Proposal/Project In-Progress Form (F5)
		Instruct student to develop Lean Model Canvas (LMC)	Appendix O
		Instruct the student to screen the FYP report with the anti-plagiarism software.	Anti-plagiarism software/system

Table 2.2 (Continued)

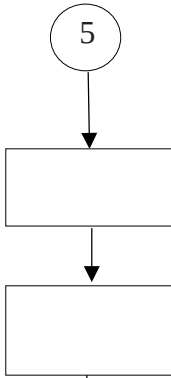
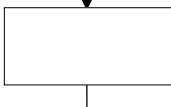
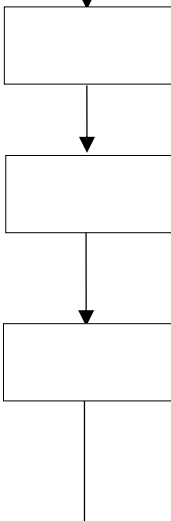
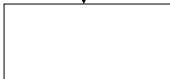
Responsibility	Workflow	Work Process	Source/Record
Project Lecturer		Evaluate the submitted LMC using LMC Evaluation Form (F13) Evaluate the project progress through presentation using the F9 form.	LMC Evaluation Form (F13) Project Progress Presentation Form (F9)
Supervisor		Review the FYP report content and format.	Appendix A-L
Project Lecturer		Update the list of supervisors and examiners. Instruct the students to submit the FYP report with the F6(b) Form. Distribute the FYP reports, with the F10 Form and the F11 Form, to the supervisors and examiners.	List of supervisors and examiners Report Submission Form (F6(b)) Final Project Presentation Form (F10) Project Report Evaluation Form (F11)
Resource Person/ FYP Coordinator		Plan and organize the FYP exhibition. Brief the students on the FYP exhibition.	

Table 2.2 (Continued)

Responsibility	Workflow	Work Process	Source/Record
Supervisor & Examiner			Final Project Presentation Form (F10)
		Evaluate the FYP presentation, report and poster using the F10 and F11 forms.	Project Evaluation Form (F11)
		Submit the forms to the Project lecturer.	FYP Report Checklist (Appendix N)
		Instruct the students to do the report amendments (if any) according to the report guidelines.	Confirmation of Correction Form(F12)
		Check the amended report and sign the F12 form.	
Supervisor			
			
			
		Sign the Supervisor Approval page in the FYP report.	
		Collect the following items from each student: • FYP report (including abstract and appendices) • Presentation slides • All files related to the project	

Project Lecturer	<pre> graph TD A[] --> B[] B --> C[] C --> D([]) </pre>	development • Poster • Raw data (if any) • System with test data (if any) • Instructions on System Setup • .apk file / .exe file (if any)	
		Record and upload the Project course marks into the RES.	Result Examination System (RES)
Project Lecturer		Submit the FYP reports to the faculty's Academic Affairs Office for archive as resource reference.	
		End	

Essentials of Computing Sciences Project Administration

2.3 Process for the Student

Table 2.3 shows the responsibilities and actions to be taken by the student during the Project Formulation and Project course.

Table 2.3 Administration Process Undertaken by the Student During the Project Formulation and Project Course

Project Formulation	
Responsibility	Action
	1. Register for the Project Formulation course. 2. Identify the project area and proposal title 3. Identify the potential project supervisor. 4. Submit a copy of the completed Mutual Acceptance Form (F1) to the Project Formulation lecturer and the supervisor. The student is asked to keep a copy for reference. The form is to be signed by both the supervisor and the student.

Student	5. Discuss the project proposal in detail with the supervisor and complete the Project Motivation Evaluation Form (F2). 6. Submit the Project Motivation Evaluation Form (F2) to the Project Formulation lecturer for review. 7. Submit Literature Review Evaluation Form (F3) and Methodology Evaluation Form (F4) to the Project Formulation lecturer for review. 8. Apply for Ethical Approval (if necessary). 9. Discuss the progress of the proposal through regular consultations with the supervisor and update the Proposal/Project In-Progress Form (F5) after each consultation. 10. Present the progress of the proposal to the Project Formulation lecturer. 11. Complete the proposal report according to the pre-specified format. 12. Screen the proposal report with the anti-plagiarism software. 13. Submit the proposal report and the F6(a) form to the Project Formulation lecturer. 14. Present the proposal to the Project Formulation lecturer, examiner and the supervisor. 15. Do the amendments on the proposal report based on the comments from the examiner and the supervisor. 16. Submit the REC forms as appropriate with the proposal. 17. Submit the amended proposal report to the Project Formulation lecturer.
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Table 2.3 (Continued)

Project	
Responsibility	Action
	18. Register for the Project course. 19. Continue regular project consultations with the supervisor and update the Proposal/Project In-Progress Form (F5) for each consultation. 20. Present the progress of the project to the supervisor and the Project lecturer. 21. Prepare the FYP report according to the report guidelines. 22. Submit Lean Model Canvas (LMC) to the Project lecturer for review.

Student	23. Screen the FYP report with the anti-plagiarism software. 24. Submit the FYP report and the Report Submission Form (F6(b)) to the Project lecturer (for the examiner and the supervisor). 25. Attend the FYP exhibition briefing conducted by the FYP coordinator. 26. Present the FYP to the: • Examiner • FYP exhibition committee members 27. Do amendments to the FYP report based on the comments from the examiner and the supervisor. 28. Submit the amended FYP report to the examiner for checking purposes and obtain the supervisor/examiner's signature on the Confirmation of Correction Form (F12). 29. Obtain the supervisor's signature on the Supervisor Approval page in the FYP report.
---------	---

Table 2.3 (Continued)

Responsibility	Action
Student	32. Submit the following items to the Project lecturer: • FYP report (including abstract and appendices) (in .pdf and .doc) • Presentation slides • Poster • All files related to the project development • Raw data (If relevant) • System with test data (if any)

2.1.2 Process Workflow for the Student

Table 2.4 shows the responsibilities, workflow, and work processes undertaken by the student during the Project Formulation and Project course.

Table 2.4 Administration Workflow for the Student During the Project Formulation and Project Course

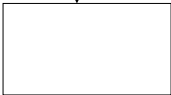
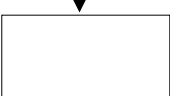
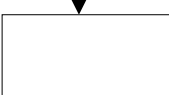
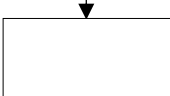
Project Formulation			
Responsibility	Workflow	Work Process	Source/Record
Student		Start	
	↓		
		Register for the Project Formulation course.	SIMS
	↓		
		Identify the project area and proposal title.	
	↓		
		Identify the potential project supervisor.	
	↓		
		Submit a copy of the F1 form to the Project Formulation lecturer and the supervisor. The student is asked to keep a copy for reference. The form is to be signed by both the supervisor and the student.	Mutual Acceptance Form (F1)
	↓		
		Discuss the project proposal in detail with the supervisor and complete the F2 form.	Project Motivation Evaluation Form (F2)
	↓		
		Submit the F2 form to the Project Formulation lecturer for review.	Project Motivation Evaluation Form (F2)
	↓		
		Submit Literature Review Evaluation Form (F3) and Methodology Evaluation Form (F4) to the Project Formulation lecturer for review.	Literature Review Evaluation Form (F3) and Methodology Evaluation Form (F4)
	↓		
		Apply for Ethical Approval (if necessary).	
	↓		
			

Table 2.4 (Continued)

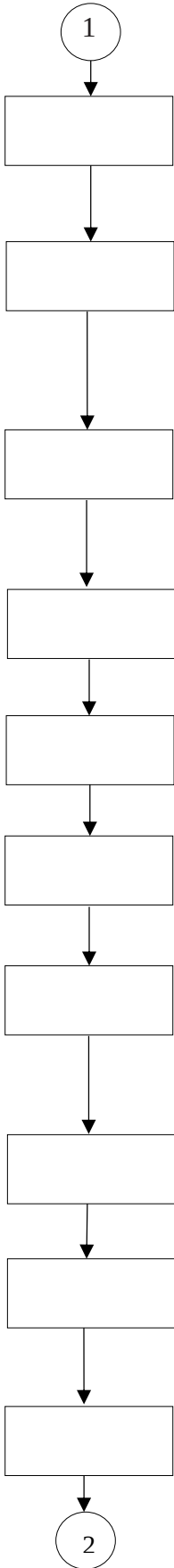
Responsibility	Workflow	Work Process	Source/Record
Student	 <pre> graph TD 1((1)) --> B1[] B1 --> B2[] B2 --> B3[] B3 --> B4[] B4 --> B5[] B5 --> B6[] B6 --> B7[] B7 --> B8[] B8 --> B9[] B9 --> B10[] B10 --> 2((2)) </pre>	Present the project proposal for evaluation. Discuss the progress of the proposal through regular consultations with the supervisor and update the F5 form after each consultation. Present the progress of the proposal to the Project Formulation lecturer. Complete the proposal report according to the pre-specified format. Screen the proposal report with the anti-plagiarism software. Submit Business Model Canvas (BMC) to the Project lecturer for review. Submit TWO (2) copies of the proposal report and the F6(a) form to the Project Formulation lecturer. Present the proposal to the Project Formulation lecturer, examiner and the supervisor. Do the amendments on the proposal report based on the comments from the examiner and the supervisor. Submit the amended proposal report to the Project Formulation lecturer.	Proposal/Project In-Progress Form (F5) Anti-plagiarism software Proposal report Report Submission Form (F6(a)) Proposal report (amended)

Table 2.4 (Continued)

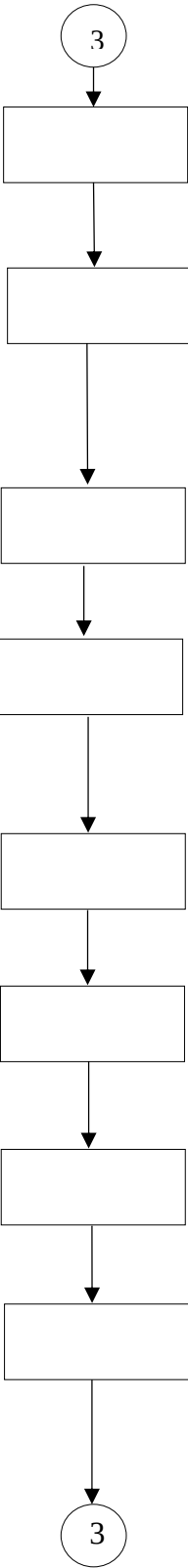
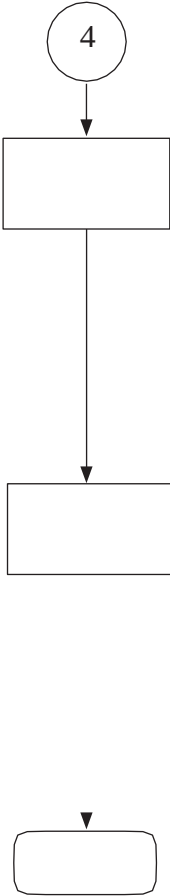
Project			
Responsibility	Workflow	Work Process	Source/Record
Student	 <pre> graph TD Start((3)) --> B1[] B1 --> B2[] B2 --> B3[] B3 --> B4[] B4 --> B5[] B5 --> B6[] B6 --> B7[] B7 --> End((3)) </pre>	Register for the Project course.	SIMS
		Continue regular project consultations with the supervisor and update the F5 form for each consultation.	Proposal/Project In-Progress Form (F5)
		Prepare Lean Model Canvas (LMC)	Appendix O
		Present the progress of the project to the supervisor and the Project lecturer.	
		Prepare the FYP report according to the report guidelines.	Appendix A – L
		Screen the FYP report with the anti-plagiarism software.	Anti-plagiarism software
		Submit TWO (2) copies of the FYP report and the F6(b) form to the Project lecturer.	Report Submission Form (F6(b))
		Attend the FYP exhibition briefing.	

Table 2.4 (Continued)

Responsibility	Workflow	Work Process	Source/Record
Student	<pre> graph TD 3((3)) --> B1[] B1 --> B2[] B2 --> B3[] B3 --> B4[] B4 --> 4((4)) </pre>	Present the FYP to the: • Examiner • FYP exhibition committee members . .	
		Complete the amendments to the FYP report based on the comments from the examiner and the supervisor.	FYP Report Checklist (Appendix N)
		Submit the amended FYP report to the examiner for checking purposes and obtain the examiner's signature on the F12 form.	Confirmation of Correction Form (F12)
		Obtain the supervisor's signature on the Supervisor Approval page in the FYP report.	

Table 2.4 (Continued)

Responsibility	Workflow	Work Process	Source/Record
Student	 <pre> graph TD A((4)) --> B[] B --> C[] C --> D([End]) </pre>	Submit the following items to the Project lecturer: • TWO (2) CD copies of the FYP report (for the FYP coordinator and the faculty's Academic Affairs Office) Each CD should contain the following items: • FYP report (including abstract and appendices) • Presentation slides • All files related to the project development • Poster • Raw data • System with test data (if any) End	Sample for CD Cover and Label (Appendix M)